QSCP Hamilton Inspection Department

Inspection Report

Site: Hamilton Plant

No.	Status	Date
QSCP-INS-2020-017	Issued	2020-04-27

External Visual Inspection and CUI Windows - ATU-100

Scope and Responsibilities

This report details the 5-yearly external visual inspection and CUI window assessment for the ATU-100. The scope follows the requirements for piping systems as defined in API 570 and API 574, ensuring the integrity of the Amine Treating Unit assets. Jennifer Walsh, acting as the Inspection Coordinator and API 570 inspector for the QSCP Hamilton Inspection Department, is responsible for the execution of this program and the verification of field data against site-specific retirement criteria.

The inspection activities were performed to validate the current condition of the piping circuits against the historical data recorded in QSCP-INS-2015-024. The assessment includes the verification of insulation integrity and the condition of the pipe surface at designated CUI windows to identify any localized degradation that could impact the remaining life of the systems.

Inspection Strategy and Reference Data

The inspection frequency and methodology are governed by QSCP-IP-ATU100 Rev 3. This inspection plan, which utilizes a Risk-Based Inspection (RBI) approach, rationalized the CMLs to 150 across 18 lines. Per the plan, RBI intervals are set at 3 years for Class 1 sour gas, regenerator overhead, and rich amine services, while a 6-year interval is applied to all other lines within the unit.

In accordance with the CUI program, all insulated lines operating within the CUI temperature range are subject to insulation window removal or profile RT every 5 years. This document serves as the follow-up to the previous 5-yearly external visual recorded in QSCP-INS-2015-024, where CUI windows were previously inspected and cleared for continued service.

Line 2-ST-101 Condition Assessment

Line 2-ST-101 is a reboiler steam service constructed from A106 Gr.B with an OD of 60.3 mm and a nominal thickness of 3.91 mm. The design limits for this circuit are 1034 kPa(g) and 186 °C. Under the site rules for Class 3 steam piping, the maximum interval for external visual inspection is 10 years, though the CUI program mandates a 5-year cycle for insulation checks.

The retirement thickness for 2-ST-101 is established at a t-min of 2.29 mm. Any thickness reading obtained during CML surveys that falls below this value requires the issuance of an NCR. The remaining life for this line is calculated by subtracting the t-min from the actual measured thickness and dividing by the corrosion rate, which is determined by the larger of the long-term or short-term rates.

Access Limitations and Deviations

During the current inspection cycle, the CUI window for line 2-ST-101 was not opened. This deviation was due to restricted physical access at the specific location, preventing the removal of the insulation cladding for a direct visual assessment of the pipe wall. This line remains subject to the 5-year CUI program requirements despite the access constraints encountered during this mobilization.

Jennifer Walsh has noted this limitation in the asset record to ensure that access is facilitated during the next available window. All other accessible CUI windows within the ATU-100 scope were inspected in accordance with the procedures outlined in QSCP-IP-ATU100 Rev 3. The external visual inspection of the remaining piping circuits was completed without further exception.

Sign-off	Name	Function	Date	Signature
Prepared	Jennifer Walsh	Inspection Coordinator, API 570 inspector	2020-04-27	_____